

Roll No. 24/SS000005

MCA-2001

Department of Computer Application
UIET, CSJM UNIVERSITY, KANPUR
Theory of Automata & Formal Language (MCA-2001)

Semester: 2nd

Time: 03:00 hrs

End Semester Examination

Session: 2024-25

Maximum Marks: 75

All sections are compulsory.

Section---A**(Short Answer Type Questions)**

All questions are compulsory. Each question carries 3 marks.

[9*3 =27]

1. (A) Discuss basic operations on Regular Expression. [3]
- (B) Construct the language generated from the given grammar: $S \rightarrow aCa, C \rightarrow aCa/b$. [3]
- (C) Discuss components of Turing Machine. [3]
- (D) Write down the conditions for the acceptance of a String by Finite Automata with example. [3]
- (E) What do you understand by NFA with null move? [3]
- (F) Discuss pigeon hole principle with suitable example. [3]
- (G) Discuss history of Automata theory. [3]
- (H) Discuss Arden's theorem. [3]
- (I) Discuss Pumping Lemma for RE. [3]

Section---B**(Long Answer Type Questions)**

Attempt any two questions. Each question carries 12 marks.

[2*12 =24]

2. a. Discuss Chomsky Hierarchy? [4]
- b. Find the grammar for the following language $L = a^n b^n c^k$, where $n \geq 0, k \geq 2$. [4]
- c. Discuss the process of conversion of an NFA to DFA with Example. [4]
3. a. Discuss at least 8 identities of Regular expression. [4]
- b. Construct a Mealy machine equivalent to the given Moore machine. [4]

Present State	Next State		O/P
	I/P = 0	I/P = 1	
$\rightarrow q_0$	q_3	q_1	0
q_1	q_1	q_2	1
q_2	q_2	q_3	0
q_3	q_3	q_3	0

- c. Discuss 2 - way Finite Automata with example in detail. [4]
 4. a. Discuss at least 4 decision problems of RE with proof. [4]
 - b. Construct the string 0100110 from the following grammar by using leftmost derivation. [4]
- $S \rightarrow 0S/1AA, \quad A \rightarrow 0/1A/0B, \quad B \rightarrow 1/0BB$
- c. Construct the string 0100110 from the following grammar by using rightmost derivation. [4]
- $S \rightarrow 0S/1AA, \quad A \rightarrow 0/1A/0B, \quad B \rightarrow 1/0BB$

Section---C
(Long Answer Type Questions)

Attempt any two questions. Each question carries 12 marks.

[2*12=24]

5. a. Convert the given Mealy machine to an equivalent Moore machine?

[6]

Present State	I/P = 0		I/P = 1	
	Next State	O/P	Next State	O/P
→ q ₀	q ₀	1	q ₁	0
q ₁	q ₃	1	q ₃	1
q ₂	q ₁	1	q ₂	1
q ₃	q ₂	0	q ₀	1

- b. Discuss the process of Minimization of Finite Automata. Also minimize the given Automata.

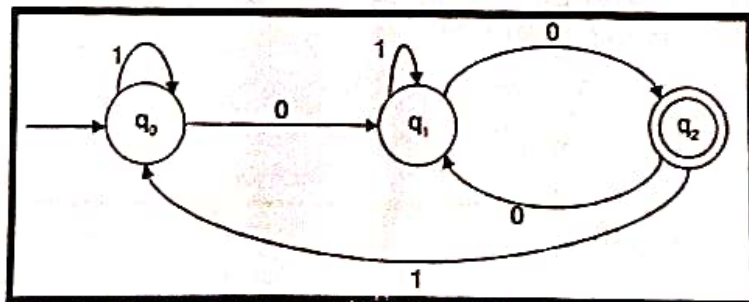
[6]

Present State	Next State	
	I/P = 0	I/P = 1
→ q ₀	q ₁	q ₂
q ₁	q ₂	q ₃
q ₂	q ₂	q ₄
⊙ q ₃	q ₃	q ₃
⊙ q ₄	q ₄	q ₄
⊙ q ₅	q ₅	q ₄

6. a. From the identities of RE, prove that $(1+100^*) + (1+100^*)(0+10^*)(0+10^*)^* = 10^*(0+10^*)^*$.
b. Construct an RE from the given FA.

[6]

[6]



7. a. Construct an FA equivalent to the RE: $L = (a + b)^*(aa + bb)(a + b)^*$

[6]

- b. What do you understand by Ambiguity in CFG?

Also check whether the given grammar is ambiguous or not. $S \rightarrow aS/AS/A, A \rightarrow AS/a$

[6]

8. a. Discuss the Simplification of CFG in detail with proper example.

[6]

- b. What do you mean by PDA? Discuss acceptance by PDA and also discuss various types of PDA in detail with examples.

[6]

**SCHOOL OF ENGINEERING AND TECHNOLOGY,
CSJM UNIVERSITY, KANPUR**

End Semester Examination-2024-25 (MCA II sem)

**Object Oriented Programming
(MCA 2002)**

Time: 3:00 hrs

Maximum marks: 75

Note: All Sections are compulsory .

Section A

Note : Attempt all questions (short questions)

[9 x 3 = 27]

- 1-a – What are the main principles of object oriented programming?
- b- What do you understand by JVM ? Explain .
- c- What are the advantages of exception handling in java?
- d- Describe the structure of a typical java program .
- e- What are different ways of creating threads?
- f- What is the purpose of package in java?
- g- What is an interface in java?
- h- What is AWT in java?
- i- How do you create a generic class in Java?

Section B

Note: Attempt any two questions from this section:

[2 x 12 = 24]

- 2-a- Why java considered to be secure and portable language?
- b- WAP in java to print a string uppercase to lowercase and lowercase to uppercase .
- c- What is an abstract class?
- 3-a – What is Layout Manager?
- b- Define the types of Layout Manager.
- c- Differentiate between classes and interfaces.
- 4-a- What is the priority of thread and how it is used in scheduling ?
- b- Discuss various AWT containers with example .
- c- How do handle multiple catch blocks for a nested try block? Explain with an example.

Section C

[2 x 12 = 24]

Note: Attempt any two questions from this section:

- 5- a- Explain Event delegation model.
- b- Explain thread life cycle and thread creation in java.
- 6-a- Write a program in java to find the area and perimeter of square and circle using interface.
- b- Explain about java Tokens with example.
- 7-a- Write a java program to create four checkbox The status of each checkbox is displayed in your output. Each time you change the state of a checkbox , the status display is updated .
- b- What is event handling in java ? Define the types of event in java .
- 8-a- What do you mean by package in java ? Explain the types of package with example .
- b- What is the purpose of generics in Java? How do you create a generic method in Java?

Roll No. 2015500005

MCA 2001

Department of Computer Application
School of Engineering and Technology, UIET
Chhatrapati Shahu Ji Maharaj University, Kanpur
End Semester Examination May -2025
Database Management System (MCA 2001)

Semester : 2nd
Time: 3 Hours

Session: 2024-25
MM: 75

Note:

- Section A - All questions are compulsory.
Section B - Attempt any two questions out of three.
Section C - Attempt any two questions out of four.

Section - A

[03 marks each]
[3 x 9 = 27]

Note: All questions are compulsory.

1.

- A. What is the purpose of a data model in DBMS?
- B. Define primary key with an example.
- C. What is the function of DCL in SQL?
- D. Name any two types of keys in relational model.
- E. What is a view in SQL?
- F. Expand ACID properties.
- G. Given the relation R (P, Q, R, S) with the following functional dependencies:
 $P \rightarrow Q$ $Q \rightarrow R$ $PR \rightarrow S$
Find all candidate keys?
- H. Define normalization.
- I. What are integrity constraints? Explain types with examples.

Section - B

[12 marks each]

Note - Attempt any two questions out of three.

- 2.
 - a. Explain the three levels of DBMS architecture with a neat diagram.
 - b. Draw an ER diagram for a library management system.
 - c. Given the relation R(A, B, C, D, E) with functional dependencies:
 $A \rightarrow B$, $B \rightarrow C$, $CD \rightarrow E$, $E \rightarrow A$
Find all candidate keys and determine the highest normal form of the relation.

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3. a. Explain two-phase locking protocol with a diagram.
 b. Compare serial and non-serial schedules with example.
 c. Given the relation $R(A, B, C, D, E, F)$ with the following FDs:
 $A \rightarrow B, C \quad C \rightarrow D \quad E \rightarrow F \quad AC \rightarrow E$
 Find all candidate keys.
4. a. Write SQL code to create a table with a primary key, then apply some aggregation function on the table.
 b. Define candidate key and foreign key with suitable examples.
 c. Given the relation $R(A, B, C, D)$ with functional dependencies:
 $AB \rightarrow C \quad C \rightarrow D \quad D \rightarrow A$
 Find all primary constraints.

Section - C

[12 marks each]

Note - Attempt any two questions out of four.

5. a. Consider the following transactions:
 $T1: \text{Read}(A); A = A - 50; \text{Write}(A);$
 $T2: \text{Read}(A); A = A * 1.1; \text{Write}(A);$
 Show a schedule involving $T1$ and $T2$ and determine if it is conflict-serializable.
 b. Discuss the purpose and techniques of recovery in DBMS.
6. a. Write and explain any four DDL commands with syntax and examples.
 b. Consider the following schedule involving three transactions:
 $T1: R(A), W(A)$
 $T2: R(A), W(A), R(B), W(B)$
 $T3: R(B), W(B)$
 Draw the precedence graph and determine if the schedule is conflict-serializable.
7. a. Consider the following schedule involving three transactions:
 $T1: R(X); W(X)$
 $T2: R(Y); W(Y)$
 $T3: R(X); W(X); R(Y); W(Y)$
 Create a schedule with interleaved operations. Determine if the schedule is view-serializable but not conflict-serializable.
 b. Given the relation $R(A, B, C, D, E)$ with functional dependencies:
 $A \rightarrow B, B \rightarrow C, CD \rightarrow E, E \rightarrow A$
 Find highest normal form.
8. a. Differentiate between DDL, DML and DCL in SQL.
 b. Explain different types of attributes used in ER modeling with examples.
~~c. Write any concurrency control technique.~~

DEPARTMENT OF COMPUTER APPLICATION
C.S.J.M. UNIVERSITY, KANPUR

Data Structures & Analysis of Algorithms (MCA-2005)

Semester: MCA 2nd Sem (Session: 2024-25)

May 2025

END SEMESTER EXAMINATION

Maximum Marks: 75

Time: 03 hrs.

All sections are compulsory.

Section A

[9 × 3 = 27]

1. Attempt all questions.

- (a) Define time complexity. Why is it important in algorithm analysis?
- (b) What is Big O notation? Provide an example.
- (c) What is the difference between a stack and a queue?
- (d) What is a priority queue and how is it implemented?
- (e) Explain the concept of dynamic memory allocation with an example.
- (f) What is the time complexity of inserting an element at the beginning of a singly linked list?
- (g) What is the maximum possible number of nodes in a binary tree at level 6?
- (h) What is a hash function? Why is it used?
- (i) Distinguish between depth-first search (DFS) and breadth-first search (BFS).

Section B

[2 × 12 = 24]

Attempt any two of the following:

2. (a) Write a C function/ algorithm to delete an element in an array.
- (b) Write a C function/ algorithm for insertion of an element in a circular queue.
- (c) Compute the time complexity of the following code segment written in C and explain the reason for the same.

.....
 for ($j = 1; j < n; j = K * j$) Statement 1;

Assuming that n is the size of input data and K is a constant.

3. (a) Convert, using algorithm, the following infix expression into its equivalent postfix expression:
 $a / b - c + d * e - a * c$
- (b) Write an algorithm to evaluate a postfix expression. Execute your algorithm to evaluate the following postfix expression as your input:
 $a b + c d + * f \uparrow$
- (c) Explain adjacency list representation of a graph with an Example.

4. (a) Show the result of inserting 3, 1, 4, 6, 9, 2, 5, 7 into an initially empty binary search tree.
- (b) What do you mean by heap? How is it different from a binary search tree? Explain with an example.
- (c) Explain different collision resolution techniques used while hashing.

Section C

Attempt **any two** of the following:

[2 × 12 = 24]

5. (a) Write C functions/ algorithms to implement a stack. Also write a function that uses your stack to reverse a string.
- (b) Write C functions/ algorithms to add a node in a doubly linked list considering all possibilities.
6. (a) Generate a binary tree for which following traversals are given:
Pre-order: + * * / A B C D E
In-order: A / B * C * D + E
- (b) The following values are to be stored in a hash table:
 25, 42, 96, 101, 102, 162, 197
 Describe how values are hashed by using division method of hashing with a table size of 7.
7. (a) Sort following integers using heap sort:
 8, 25, 16, 21, 45, 19, 17, 67, 52, 69, 70.
- (b) Explain Prim's algorithm of minimum spanning tree with suitable example.
8. (a) How AVL tree is useful over a binary search tree? Demonstrate AVL rotations when the nodes with keys 10, 20, 32, 35, 25, 27 are inserted in an AVL tree.
- (b) Demonstrate the insertion of the key (1, 2, 3, 4, 8, 7, 6, 5, 9, 10, 11, 12, 16, 15, 14, 13) in a B-tree of order 5.

School of Engineering & Technology, C.S.J.M. University

End Semester Examination - May, 2025

M.C.A. IInd Semester

MCA – 2003 (Operating System)

M.M. 75

Time: 3 hr.

Section A (This Section is compulsory.)

(9x3=27 marks)

Que. 1)

- What is 50-percent rule?
- What do you mean by system calls?
- What is the use of job queues, ready queue, and device queues?
- What is co-operative process?
- What is preemptive and non-preemptive scheduling?
- What is critical section problem?
- What is a resource-allocation graph?
- What do you mean by first fit, best fit and worst fit?
- Define the virtual memory? What are its advantages?

Section 'B' (Attempt any two questions from this section)

(12x2=24 marks)

- Que. 2) A) Explain process control box in detail.
B) What do you understand by demand paging? Explain pure demand paging.
C) What do you understand by a kernel? Explain its types and differentiate them.

- Que.3) A) Define deadlock avoidance. How can we detect and recover from this? Explain in details.
B) What is starvation? When could it happen? What are the methods to stop it?
C) Distinguish between internal and external fragmentation.

- Que.4) A) What is the purpose of system programs/system calls?
B) Explain Belady's anomaly.
C) What is asymmetric multiprocessing? Discuss the difference between symmetric and asymmetric multiprocessing.

Section C (Attempt any two questions from this section)

(12x2=24 marks)

- Que.5) Consider the following snapshot of a system:

	Allocation	Max	Available
	A B C D	A B C D	A B C D
T0	3 1 4 1	6 4 7 3	2 2 2 4
T1	2 1 0 2	4 2 3 2	
T2	2 4 1 3	2 5 3 3	
T3	4 1 1 0	6 3 3 2	
T4	2 2 2 1	5 6 7 5	

Answer the following questions using the banker's algorithm:

- A. Illustrate that the system is in a safe state by demonstrating an order in which the threads may complete.
- B. If a request from thread T_4 arrives for (2, 2, 2, 4), can the request be granted immediately?
- C. If a request from thread T_2 arrives for (0, 1, 1, 0), can the request be granted immediately?
- D. If a request from thread T_3 arrives for (2, 2, 1, 2), can the request be granted immediately?

Que.6) The following processes are being scheduled using a pre-emptive, priority-based, round-robin scheduling algorithm.

PROCESS	PRIORITY	BURST	ARRIVAL
P1	8	15	0
P2	3	20	0
P3	4	20	20
P4	4	20	25
P5	5	5	45
P6	5	15	55

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. The scheduler will execute the highest priority process for processes with the same priority; a round-robin scheduler will be used with a time quantum of 10 units. If a process is pre-empted by a higher-priority process, the pre-empted process is placed at the end of the queue.

- A. Show the scheduling order of the processes using a Gantt chart.
- B. What is the turnaround time for each process?
- C. What is the waiting time for each process?

Que.7) Consider the following page reference string:

7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1.

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms?

- A) LRU Replacement
- B) FIFO Replacement
- C) OPTIMAL Replacement

Que.8)

Discuss deadlock detection. What is Wait-For graph? Also explain in detail various methods of recovery from deadlock? How do we choose victim process?